

Probabilistic Relation-Changing Operators

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Abstract. Relation-changing logics usually deal with dynamic phenomena over discrete models. In order to describe their properties and reason in this setting, different approaches were proposed, such as syntactical ones – where the model can be changed due to the use of some logical operator in the language – or the semantical ones – where the change in the system is parameterized by the model itself. The relation-changing paradigm has been studied and extended in the last years by considering, for instance, paraconsistent and fuzzy proposals. In this work, we aim to provide relation-changing logics over probabilistic transition systems. Precisely, we propose two relation-changing operators and explore how they affect the semantics, for instance by investigating adequate notions of probabilistic bisimulation.

Keywords: Relation-Changing · Probabilistic Logic · Probabilistic Kripke Model · Bisimulation · Sabotage Logic · Bridge Logic

1 Introduction

There are many kinds of probabilistic models to describe a probabilistic transition system. For instance, structures such as Markov Chains, a well-known example of the wider class of probabilistic automata (see e.g. [17]) play this role. Together with these models, we can use logical languages to describe features or reason about them. Different languages can be employed, such as the ones based on temporal logic (used, for instance, in PRISM to describe properties of Markov chains [13] and in [12] to describe knowledge) and probabilistic modal logics (see e.g. [1]), designed to reason about probabilistic Kripke structures.

A different concept associated with transition systems is that of relation-changing systems. In general, relation-changing systems are discrete state transition models whose accessibility relation may change after some event, such as crossing an edge. The exact way this configuration change is produced can vary from a semantical to a syntactical cause. For instance, we can consider models such as switch graphs, where the relation changes are encoded by the structure of the model itself [11]. In these cases, we can explore the model using classical

modal logic, without causing any change directly. Instead, updates are found as the result of crossing specific edges, according to the model specification. Another approach consists on considering a regular Kripke structure and explore it with a logic which embeds proper relation-changing operators on their syntax. These languages, such as sabotage and bridge logic [2,3,6], contain specific operators that while evaluating a formula, they cause the remaining subformula to be evaluated in a different model. In such a model, the accessibility relation may be different from the original one.

During the last years, the relation-changing paradigm has been implemented in different classes of models such as paraconsistent [5], weighted [10], fuzzy [7] and labeled [4,9] models. Also, some applications and computational implementations have also been developed. For instance, [10] presents a specific tool to deal with probabilistic cases, by considering very specific weights. Also, in [14], the same approach is employed in a practical context, showing its applicability for real-life problems. Other case-studies have been proposed, for instance, in [8,16]. This work aims to develop the study of this area by studying the introduction of relation-changing operators to probabilistic contexts, instead of considering preconceived models with limited expressiveness. While this has been done to models with fuzzy logic [7], probabilistic setting is not derivable as a particular case, since additional conditions must be imposed over the relation-changing process, in order to preserve the properties of a probabilistic model.

In this article we present a probabilistic approach for relation-changing logics, and study associated probabilistic bisimulation notions. In Section 2 we introduce the syntax, semantics and bisimulation notion for the basic probabilistic modal logic we consider. Section 3 is devoted to exploring probabilistic versions of sabotage and bridge modalities. Therein, we present their associated notions of bisimulations and show their respective invariance theorems. We finish in Section 4 with some remarks and future lines of work.

2 Probabilistic Modal Logic

2.1 Syntax and Semantics

In this section we present some preliminary notions over probabilistic logics. In what follows, let **Prop** be a countable set of atomic propositions. Also, for every relation E , we use the abbreviation $E[U] = \{v \mid (u, v) \in E \text{ for some } u \in U\}$.

Definition 1. A probabilistic Kripke model is a tuple $\mathcal{M} = (W, P, V)$ where:

- W is a finite set of states or worlds;
- $P : W \times W \rightarrow [0, 1]$ is the transition probability matrix where for all $w \in W$,

$$\sum_{w' \in W} P(w, w') = 1;$$
- $V : \text{Prop} \rightarrow 2^W$ is a function that assigns to each $p \in \text{Prop}$ the set of states in which p holds.

We adopt the probabilistic definition of modal logic in [1]. Notice that the definition of truth is *crisp*, i.e., it is either true or false.

Definition 2 (Formulas). *The set of formulas is defined recursively as:*

$$\varphi, \psi ::= p \mid \neg\varphi \mid \varphi \vee \psi \mid \Diamond_{\rho}\varphi,$$

where $p \in \text{Prop}$ and $\rho \in [0, 1]$.

Definition 3 (Satisfaction). *Let $\mathcal{M} = (W, P, V)$ be a probabilistic Kripke structure, and let w be a state in W . For any probabilistic modal logic formula φ , we define the satisfaction relation $\models$ recursively as follows:*

$$\begin{aligned} \mathcal{M}, w \models p & \quad \text{iff } w \in V(p) \\ \mathcal{M}, w \models \neg\varphi & \quad \text{iff } \mathcal{M}, w \not\models \varphi \\ \mathcal{M}, w \models \varphi \vee \psi & \quad \text{iff } \mathcal{M}, w \models \varphi \text{ or } \mathcal{M}, w \models \psi \\ \mathcal{M}, w \models \Diamond_{\rho}\varphi & \quad \text{iff } \sum_{\{v \mid \mathcal{M}, v \models \varphi\}} P(w, v) \geq \rho. \end{aligned}$$

Example 1. To illustrate how the satisfaction is evaluated on a probabilistic Kripke model, consider the structure depicted in Fig. 1.

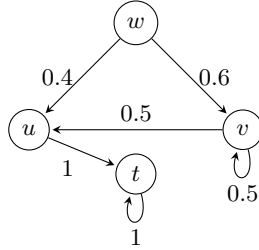


Fig. 1. Example of probabilistic Kripke model.

We define a model $\mathcal{M} = (W, P, V)$, where $W = \{u, v, w, t\}$, P is the transition probability matrix depicted in Table 1 and $V : \{p, q\} \rightarrow 2^W$ is defined by $V(p) = \{u, w, t\}$ and $V(q) = \{v, t\}$.

As example, let us evaluate the formula $\Diamond_{0.5}(p \vee \Diamond_{0.8} q)$ at the state w :

$$\mathcal{M}, w \models \Diamond_{0.5}(p \vee \Diamond_{0.8} q) \quad \text{iff} \quad \sum_{\{x \in W \mid \mathcal{M}, x \models p \vee \Diamond_{0.8} q\}} P(w, x) \geq 0.5.$$

Thus, for each state $x \in W$, we need to check whether $\mathcal{M}, x \models p \vee \Diamond_{0.8} q$, that holds if and only if $\mathcal{M}, x \models p$ or $\mathcal{M}, x \models \Diamond_{0.8} q$. Since $V(p) = \{u, w, t\}$, we only need to check the case $x = v$:

Table 1. Transition probability matrix of model depicted in Fig. 1.

	u	v	w	t
u				1
v	0.5	0.5		
w	0.4	0.6		
t				1

$\mathcal{M}, v \models \Diamond_{0.8} q$ iff $\sum_{\{y \in W \mid \mathcal{M}, y \models q\}} P(v, y) \geq 0.8$. According to the semantics, this is the case iff $P(v, v) + P(v, t) \geq 0.8$. But then we get $0.5 \geq 0.8$, that is false.

Hence, $\sum_{\{x \in W \mid \mathcal{M}, x \models p \vee \Diamond_{0.8} q\}} P(w, x) \geq 0.5$ holds iff $P(w, w) + P(w, u) + P(w, t) \geq 0.5$, iff $0.4 \geq 0.5$, which is false.

$\therefore \mathcal{M}, w \not\models \Diamond_{0.5}(p \vee \Diamond_{0.8} q)$.

2.2 Bisimulation

We now introduce a definition of bisimulation for the basic probabilistic case. While the classical definition usually defines bisimulation as an equivalence relation over the states of a probabilistic Kripke model, we introduce a version here to relate two different probabilistic Kripke models.

In [15], Panangaden studies probabilistic structures with labeled edges but without propositions. In that paper, the author introduces the definition of dynamic relation based on the work of Breugel & Worrell [19,20], which does not require the relation to be an equivalence relation. Moreover, he proves that a dynamic relation implies the existence of a bisimulation relation. There, he was also able to prove the invariance and Hennessy-Milner theorem for finite models. Despite that paper considers a probabilistic Kripke structure with labels, we use it as the basis for our definition within a context that also considers propositions.

Definition 4 (Bisimulation). Let $\mathcal{M} = (W, P, V)$ and $\mathcal{M}' = (W', P', V')$ be two probabilistic Kripke models and $E \subseteq W \times W'$. We say that a nonempty relation E is a bisimulation from $\mathcal{M}$ to $\mathcal{M}'$ if for every $w \in W$ and $w' \in W'$ such that wEw' , we have

atoms: for any $p \in \text{Prop}$, $w \in V(p)$ iff $w' \in V'(p)$,
zigzag_{prob}: for any $U \subseteq W$, $\sum_{u \in U} P(w, u) \leq \sum_{u' \in E[U]} P'(w', u')$

With this definition at hand, we are able to prove the following characterization result, usually called the *invariance under bisimulation theorem*.

Theorem 1. Let $\mathcal{M} = (W, P, V)$ and $\mathcal{M}' = (W', P', V')$ be two probabilistic Kripke models and $E \subseteq W \times W'$ a bisimulation from $\mathcal{M}$ to $\mathcal{M}'$. Then for any $(w, w') \in E$ and any formula φ we have

$$\mathcal{M}, w \models \varphi \text{ iff } \mathcal{M}', w' \models \varphi$$

Proof. The proof is obtained as a fragment of the proof of Theorem 2, that will be presented later on the paper. $\square$

3 Relation changing operators

This section presents our main contribution, which is defining novel logical languages which embed a relation-changing operator, interpreted over probabilistic Kripke structures. The languages extend those in e.g. [18,1]. The ‘update’ modality (written generally $\blacklozenge_\sigma$ here) will have two possible interpretations. First, as a sabotage-like modality (see [6,2,3]) in which the probability is subtracted from the edge accessing a chosen successor of the evaluation point, while it is transferred to another edge. Second, we introduce a bridge-like modality [6,2], in which the probability is instead taken from some edge to be added to the edge accessing the chosen successor.

The language we use is given by the following BNF:

$$\varphi, \psi ::= p \mid \neg\varphi \mid \varphi \vee \psi \mid \blacklozenge_\rho\varphi \mid \blacklozenge_\sigma\varphi,$$

where $p \in \mathbf{Prop}$, $\rho \in [0, 1]$ and $\sigma \in]0, 1[$. We denote the full set of formulas generated by the BNF above by $\mathbf{Form}$. In the following, if the context is clear, we will use both $\blacklozenge_\rho$ and $\blacklozenge_\rho$.

The idea of changing a probabilistic relation implies that the transition probability matrix may change. We define $Prob(W)$, the set of all transition probability matrices over the set W as:

$$Prob(W) = \{P \in [0, 1]^{W \times W} : \forall w \in W, \sum_{w' \in W} P(w, w') = 1\}.$$

3.1 Probabilistic Sabotage Operator

Let us start by introducing the semantics of a local sabotage modality. In [6,2], a local sabotage is interpreted as moving the evaluation of the formula to a successor of the current point, while removing the traversed edge. In the graded/fuzzy case [7], edges have associated values, so the sabotage is carried out instead by reducing the value of the traversed edge. Here, we will follow a similar approach, but maintaining the properties of a probabilistic Kripke model.

Definition 5. Let $\mathcal{M} = (W, P, V)$ be a probabilistic Kripke structure, and let w be a state in W . For any formula $\varphi \in \mathbf{Form}$, we define the satisfaction relation $\models$ recursively as follows:

$$\begin{aligned} \mathcal{M}, w \models p & \quad \text{iff } w \in V(p) \\ \mathcal{M}, w \models \neg\varphi & \quad \text{iff } \mathcal{M}, w \not\models \varphi \\ \mathcal{M}, w \models \varphi \vee \psi & \quad \text{iff } \mathcal{M}, w \models \varphi \text{ or } \mathcal{M}, w \models \psi \\ \mathcal{M}, w \models \blacklozenge_\rho\varphi & \quad \text{iff } \sum_{\{v \mid \mathcal{M}, v \models \varphi\}} P(w, v) \geq \rho \\ \mathcal{M}, w \models \blacklozenge_\sigma\varphi & \quad \text{iff } \text{exists } v \in W \text{ s.t. } \rho \leq P(w, v) \neq 1, \text{ and } \mathcal{M}_\rho^{-(w, v)}, v \models \varphi, \end{aligned}$$

where $\mathcal{M}_\rho^{-(w,v)} = (W, P_\rho^{-(w,v)}, V)$, with:

- $P_\rho^{-(w,v)}(w, v) = P(w, v) - \rho$;
- $P_\rho^{-(w,v)}(w, t) = P(w, t) + \frac{\rho \cdot P(w, t)}{1 - P(w, v)}$, if $t \neq v$;
- $P_\rho^{-(w,v)}(s, t) = P(s, t)$, if $s \neq w$.

Note that $\mathcal{M}_\rho^{-(w,v)}$ is a well-formed model, since it is updated in the context of the precondition $P(w, v) \geq \rho$ and, according to its definition, $P_\rho^{-(w,v)} \in \text{Prob}(W)$.

The semantic interpretation of the sabotage operator describes a scenario in which the probability of the event represented by an edge is reduced. When that is the case, the probability of the remaining ones (that also leave from the same origin state) is increased, in order to guarantee that the probability of all leaving edges sums to 1. For each of these edges, the increase is proportional to their pre-existing probability, in order to preserve the likelihood of each one being crossed. This is illustrated in Example 2.

Example 2. Let us consider the model (W, P, V) introduced in Example 1, and check if w satisfies $\blacklozenge_{0.4}p$. By the semantics we have $\mathcal{M}, w \models \blacklozenge_{0.4}p$ iff $\exists x \in W$ s.t. $0.4 \leq P(w, x) \neq 1$, and $\mathcal{M}_{0.4}^{-(w,x)}, x \models p$.

$P(w, x) \geq 0.4$ implies $x = u$ or $x = v$. Although $v \notin V(p)$, the formula is satisfied because $\mathcal{M}_{0.4}^{-(w,u)}, u \models p$. Moreover, the model $\mathcal{M}_{0.4}^{-(w,u)} = (W, P_{0.4}^{-(w,u)}, V)$ is illustrated in Fig. 2.

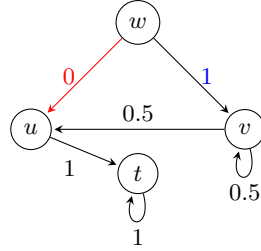


Fig. 2. The probabilistic Kripke model $(W, P_{0.4}^{-(w,u)}, V)$.

3.2 Probabilistic Bridge Operator

Now it is time to move to the bridge case, generalizing [6,2], and being more specific that in the graded case from [7]. The idea is that if there no way to access some point with a certain probability value, the evaluation is moved to such a point, and the value is added to the traversed edge. This is similar to the graded

case, but here we need to be careful to maintain the probability distribution. This is achieved by proportionally transferring the value from other edges to the intended one.

Definition 6. *The satisfiability relation $\models$ is defined as:*

$$\begin{aligned}
 \mathcal{M}, w &\models p && \text{iff } w \in V(p) \\
 \mathcal{M}, w &\models \neg\varphi && \text{iff } \mathcal{M}, w \not\models \varphi \\
 \mathcal{M}, w &\models \varphi \vee \psi && \text{iff } \mathcal{M}, w \models \varphi \text{ or } \mathcal{M}, w \models \psi \\
 \mathcal{M}, w &\models \Diamond_\rho \varphi && \text{iff } \sum_{\{v \mid \mathcal{M}, v \models \varphi\}} P(w, v) \geq \rho \\
 \mathcal{M}, w &\models \blacklozenge_\rho \varphi && \text{iff exists } v \in W \text{ s.t. } P(w, v) + \rho \leq 1, \text{ and } \mathcal{M}_\rho^{+(w,v)}, v \models \varphi,
 \end{aligned}$$

where $\mathcal{M}_\rho^{+(w,v)} = (W, P_\rho^{+(w,v)}, V)$, with:

$$\begin{aligned}
 - P_\rho^{+(w,v)}(w, v) &= P(w, v) + \rho; \\
 - P_\rho^{+(w,v)}(w, t) &= P(w, t) - \frac{\rho \cdot P(w, t)}{1 - P(w, v)}, \text{ if } t \neq v; \\
 - P_\rho^{+(w,v)}(s, t) &= P(s, t), \text{ if } s \neq w.
 \end{aligned}$$

Again, note that $\mathcal{M}_\rho^{+(w,v)}$ is a well-formed model.

The semantics of the bridge operator depicts a case where either the probability of an event to occur increases or some new transition is possible. Thus, the probability assigned to the other edges that also leave from the same origin state must decrease. As in the case of the “sabotage”, this reduction is obtained in a proportional way, based on the previous value of each edge probability. This is illustrated in Example 3.

Example 3. Let us recall the model (W, P, V) from Example 1 and check if v satisfies $\blacklozenge_{0.4} q$. By the semantics we have $\mathcal{M}, v \models \blacklozenge_{0.4} q$ iff $\exists x \in W$ s.t. $0.4 \leq P(v, x) \neq 1$, and $\mathcal{M}_{0.4}^{+(v,x)}, x \models q$.

$P(v, x) \leq 0.4$ implies $x = w$ or $x = t$. Although $w \notin V(p)$, the formula is satisfied because $\mathcal{M}_{0.4}^{+(v,t)}, t \models p$. Moreover, the model $\mathcal{M}_{0.4}^{+(v,t)} = (W, P_{0.4}^{+(v,t)}, V)$ is illustrated in Fig. 3.

3.3 Bisimulation

Based on the Def. 4, we introduce a notion of bisimulation for the probabilistic language obtained by adding either the sabotage or bridge operator.

Definition 7 (Bisimulation). *Let $\mathcal{M} = (W, P, V)$ and $\mathcal{M}' = (W', P', V')$ be two probabilistic Kripke models. We say that a non-empty relation $E \subseteq (W \times \text{Prob}(W)) \times (W' \times \text{Prob}(W'))$ is a ρ -sabotage (resp. ρ -bridge) bisimulation from $\mathcal{M}$ to $\mathcal{M}'$ if, for every pair $((w, P), (w', P')) \in E$, the following holds:*

atom: $w \in V(p)$ iff $w' \in V'(p)$, for all $p \in \text{Prop}$;

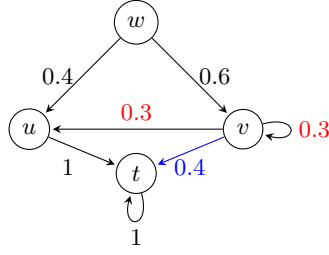


Fig. 3. The probabilistic Kripke model $(W, P_{0.4}^{+(v,t)}, V)$.

- zigzag_{prob}**: for any $U \subseteq W$, $\sum_{u \in U} P(w, u) \leq \sum_{(u', P') \in \bigcup_{u \in U} E[(u, P)]} P'(w', u')$;
- r-zig_{prob}**: for any $u \in W$ s.t. $1 \neq P(w, u) > \rho$ (resp. $P(w, u) + \rho < 1$), exists $u' \in W'$ such that:
1. $P(w, u) \leq P'(w', u') < 1$ (resp. $P(w, u) \geq P'(w', u')$), and
 2. $(u, P_{\rho}^{-(w,u)})E(u', P_{\rho}'^{-(w',u')})$ (resp. $(u, P_{\rho}^{+(w,u)})E(u', P_{\rho}'^{+(w',u')})$);
- r-zag_{prob}**: for any $u' \in W'$ s.t. $1 \neq P'(w', u') > \rho$ (resp. $P'(w', u') + \rho < 1$), exists $u \in W$ such that:
1. $P'(w', u') \leq P(w, u) < 1$ (resp. $P'(w', u') \geq P(w, u)$ and
 2. $(u, P_{\rho}^{-(w,u)})E(u', P_{\rho}'^{-(w',u')})$ (resp. $(u, P_{\rho}^{+(w,u)})E(u', P_{\rho}'^{+(w',u')})$).

Moreover, for ρ -bridge bisimulation, we also impose that for all $w \in W, w' \in W'$, $E[\{(w, P)\}] \neq \emptyset \neq E^{-1}[\{(w', P')\}]$.

We say that E is a *sabotage bisimulation* (resp. *bridge bisimulation*) if it is a ρ -sabotage (resp. ρ -bridge) bisimulation for every $\rho \in [0, 1]$. Moreover, if E is a bisimulation and $(w, P)E(w', P')$, we say that w and w' are two *sabotage* (resp. *bridge*) *bisimilar states*.

We note that our definition of sabotage bisimulation does not fully generalize the classical one. However, this is indeed caused by the semantical interpretation of the new operator $\Diamond_{\rho}$ that can distinguish two probabilistic Kripke models such as the ones illustrated in Fig. 4, which are bisimilar according to the classical definition. Thus, contrary to the classical case, we need to consider both **r-zig_{prob}** and **r-zag_{prob}** conditions. As it is possible to note in Fig. 4, the **r-zig_{prob}** condition is verified from left to right model, but not in the opposite direction.

Theorem 2 (Invariance). *Let $\mathcal{M} = (W, P, V)$ and $\mathcal{M}' = (W', P', V')$ be two probabilistic Kripke models and let $E \subseteq ((W \times \text{Prob}(W)) \times (W' \times \text{Prob}(W')))$ be a sabotage (resp. bridge) bisimulation. Then, $(w, P)E(w', P')$ implies $\mathcal{M}, w \models \varphi$ iff $\mathcal{M}, w' \models \varphi$, for every $\varphi \in \text{Form}$, under sabotage (resp. bridge) semantics.*

Proof. We prove the theorem by induction over the structure of formulas. We firstly consider the sabotage semantics.

Base case:

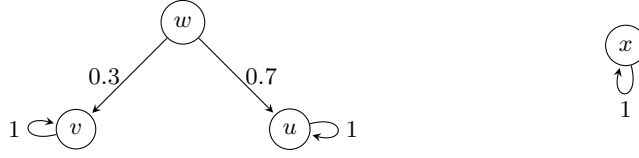


Fig. 4. Two model whose states are bisimilar but not sabotage/bridge bisimilar.

- If $\varphi \in \mathbf{Prop}$, then the result follows from **atom** from Def. 7.

Inductive cases:

- If $\varphi = \neg\psi$, then $\mathcal{M}, w \models \neg\psi$ iff $\mathcal{M}, w \not\models \psi$. By IH, $\mathcal{M}', w' \not\models \psi$, and then $\mathcal{M}', w' \models \neg\psi$.
- If $\varphi = \psi \vee \chi$, then $\mathcal{M}, w \models \psi \vee \chi$ iff $\mathcal{M}, w \models \psi$ or $\mathcal{M}, w \models \chi$. By IH, we get $\mathcal{M}', w' \models \psi$ or $\mathcal{M}', w' \models \chi$, hence $\mathcal{M}', w' \models \psi \vee \chi$.
- If $\varphi = \Diamond_\rho \psi$, then $\mathcal{M}, w \models \Diamond_\rho \psi$ iff $\sum_{(v \mid \mathcal{M}, v \models \varphi)} P(w, v) \geq \rho$. Note that (by Def. 7) $\sum_{\{v \mid \mathcal{M}, v \models \psi\}} P(w, v) \leq \sum_{\{v' \mid (v', P') \in \bigcup_{\{v \mid \mathcal{M}, v \models \psi\}} E[(v, P)]\}} P'(w', v') \leq \sum_{\{v' \mid \mathcal{M}', v' \models \psi\}} P'(w', v')$ (since $(v, P)E(v', P')$, and by IH). Moreover, by using the same reasoning we can conclude that $\sum_{\{v \mid \mathcal{M}, v \models \neg\psi\}} P(w, v) \leq \sum_{\{v' \mid \mathcal{M}', v' \models \neg\psi\}} P'(w', v')$. Since for all $w \in W$, $\sum_{v \in W} P(w, v) = 1$ and for all $w' \in W'$, $\sum_{v' \in W'} P'(w', v') = 1$, we get that $\sum_{\{v \mid \mathcal{M}, v \models \psi\}} P(w, v) = \sum_{\{v' \mid \mathcal{M}', v' \models \psi\}} P'(w', v')$, because every $w \in W \cup W'$ satisfies either ψ or $\neg\psi$. Hence, we can conclude $\mathcal{M}, w \models \Diamond_\rho \psi$ iff $\mathcal{M}', w' \models \Diamond_\rho \psi$.
- If $\varphi = \Diamond_\rho \psi$, then $\mathcal{M}, w \models \Diamond_\rho \psi$ iff exists $v \in W$ s.t. $\rho \leq P(w, v) < 1$, and $\mathcal{M}_\rho^{-(w, v)}, v \models \psi$. From **r-zig_{prob}** in Def. 7, we conclude that exists $v' \in W'$ s.t. $P(w, v) \leq P'(w', v')$ and $(v, P_\rho^{-(w, v)})E(v', P_\rho^{-(w', v')})$. Thus, by IH, exists $v' \in W'$, s.t. $\rho \leq P'(w', v') < 1$, and $\mathcal{M}'_\rho^{-(w', v')}, v' \models \psi$, and therefore $\mathcal{M}', w' \models \Diamond_\rho \psi$.

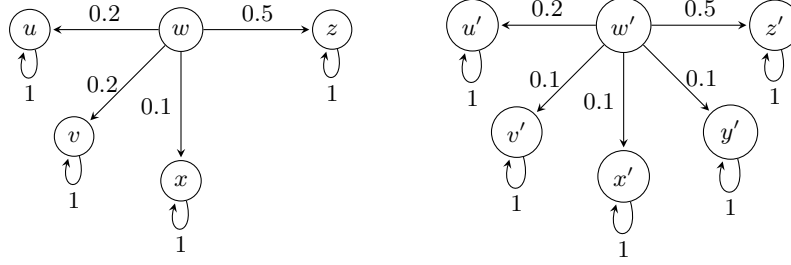
The reciprocal implication is proved analogously, using **r-zag_{prob}**.

Finally, for bridge semantics, the result is obtained using the analogous reasoning. Here we note that the additional condition that for all $w \in W$, $w' \in W'$, $E[\{(w, P)\}] \neq \emptyset \neq E^{-1}[\{(w', P')\}]$ ensures that it is not possible to create a bridge to a new state that was previously not bisimilar to any other. This is not required for sabotage because the probability assigned to edges with probability of 0 is never changed. $\square$

Example 4. Let $\mathbf{Prop} = \{p\}$ and consider the two probabilistic Kripke models, as shown in Fig. 5. We denote the model on the left by $\mathcal{M} = (W, P, V)$ and the one on the right by $\mathcal{M}' = (W', P', V')$, with $V(p) = \{u, v, x\}$ and $V'(p) = \{u', v', x', y'\}$.

It is possible to verify that the following relation is a sabotage-bisimulation:

$$\begin{aligned} E = & \{((w, P), (w', P')), ((z, P), (z', P'))\} \\ & \cup \{(t, P) \times (t', P') : t \in \{u, v, x\} \text{ and } t' \in \{u', v', x', y'\}\} \end{aligned}$$

**Fig. 5.** Example of models with sabotage/bridge-bisimilar states.

$$\begin{aligned}
& \cup \{(z, P_\rho^{-(w,z)}) \times (z', P_{\rho'}^{-(w',z')}) : \rho \in]0, 0.5]\} \\
& \cup \{(t, P_\rho^{-(w,t)}) \times (u', P_{\rho'}^{-(w',u')}) : t \in \{u, v\} \text{ and } \rho \in]0, 0.2]\} \\
& \cup \{(x, P_\rho^{-(w,x)}) \times (t', P_{\rho'}^{-(w',t')}) : t' \in \{v', x', y'\} \text{ and } \rho \in]0, 0.1]\}
\end{aligned}$$

$\therefore w$ is bisimilar to w' , z is bisimilar to z' and u, v, x are bisimilar to u', v', x', y' .

4 Conclusion and future work

In this paper, we extend the paradigm of relation-changing logics to probabilistic transition systems. While traditional relation-changing logics address dynamic phenomena over discrete models, we adapt this framework to settings where transitions are governed by probabilities. Specifically, we introduce two novel relation-changing operators tailored to the probabilistic case and analyze their impact on the underlying semantics. To support reasoning in this setting, we also investigate appropriate notions of probabilistic bisimulation, showing how these operators interact with key behavioral equivalences. This work thus opens a new direction in the study of relation-changing logics, complementing existing proposals in paraconsistent, fuzzy, and graded contexts [5, 7].

This work is closely related to graded relation updates from [7], where classical relation-changing operators are extended to graded (fuzzy) relational structures. While that approach captures degree-based updates of accessibility, our proposal focuses on distributional updates over probabilistic transition systems, thus providing a complementary perspective on extending relation-changing logics beyond the classical setting.

As future work, we aim to develop model checking tools for relation-changing contexts. Although in a different context, rPRISM [10] – a tool which was integrated with PRISM – can be a starting point. We also would like to study the decidability status/complexity of the model checking problem and propose an automatic model-checker that integrates probabilistic sabotage and bridge operators.

Also, we would like to explore different semantics for relation-changing operators. In this approach, the changing operators are defined in such a way that

we move to a new accessible state (with an appropriate probability) after the change of the model. It would be interesting to study the case where we simply update the model without moving from the current state. A global version of these operators – where changes are not made locally but anywhere in the transition probability matrix of the model – would also be an interesting topic to explore.

Finally, we intend to explore the existence of a Hennessy-Milner theorem for finite models concerning our definition of bisimulation in the probabilistic setting, as well as in the model changing case.

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References

1. Alessandro Aldini, Gianluca Curzi, Pierluigi Graziani, and Mirko Tagliaferri. A probabilistic modal logic for context-aware trust based on evidence. *International Journal of Approximate Reasoning*, 169:109167, 2024.
2. Carlos Areces, Raul Fervari, and Guillaume Hoffmann. Relation-changing modal operators. *Logic Journal of the IGPL*, 23(4):601–627, 2015.
3. Guillaume Aucher, Johan van Benthem, and Davide Grossi. Modal logics of sabotage revisited. *Journal of Logic and Computation*, 28(2):269–303, 2018.
4. Suene Campos, Daniel Figueiredo, Manuel A. Martins, and Regivan Santiago. Labeled fuzzy reactive graphs. *Fuzzy Sets and Systems*, page 109320, 2025.
5. Diana Costa, Daniel Figueiredo, and Manuel A. Martins. Relation-changing models meet paraconsistency. *Journal of Logical and Algebraic Methods in Programming*, 133:100870, 2023.
6. Raul Fervari. *Relation-Changing Modal Logics*. PhD thesis, Universidad Nacional de Córdoba, 2014.
7. Raul Fervari, Daniel Figueiredo, and Manuel A. Martins. Graded relation updates in modal logic. In *Logic, Language, Information, and Computation - 31st International Workshop, WoLLIC 2025*, volume 15942 of *Lecture Notes in Computer Science*, pages 278–292. Springer, 2025.
8. Daniel Figueiredo and Luís Soares Barbosa. Reactive models for biological regulatory networks. In *International Symposium on Molecular Logic and Computational Synthetic Biology*, pages 74–88. Springer, 2018.
9. Daniel Figueiredo, Manuel A. Martins, and Luís S Barbosa. A note on reactive transitions and reo connectors. In *It’s All About Coordination: Essays to Celebrate the Lifelong Scientific Achievements of Farhad Arbab*, pages 57–67. Springer, 2018.

10. Daniel Figueiredo, Eugénio Rocha, Manuel A. Martins, and Madalena Chaves. rPrism – a software for reactive weighted state transition models. In *Hybrid Systems Biology*, pages 165–174, Cham, 2019. Springer International Publishing.
11. Dov M Gabbay and Sérgio. Marcelino. Global view on reactivity: switch graphs and their logics. *Annals of Mathematics and Artificial Intelligence*, 66(1-4):131–162, 2012.
12. Joseph Y Halpern and Mark R Tuttle. Knowledge, probability, and adversaries. *Journal of the ACM (JACM)*, 40(4):917–960, 1993.
13. Marta Kwiatkowska, Gethin Norman, and David Parker. *Stochastic model checking*, pages 220–270. Springer, 2007.
14. Diogo Mendes, Daniel Figueiredo, Carlos Alves, Ana Penedones, Beatriz Costa, and Francisco Batel-Marques. Impact of the covid-19 pandemic on cancer screenings in portugal. *Cancer Epidemiology*, 88:102496, 2024.
15. Prakash Panangaden. Probabilistic bisimulation. In Davide Sangiorgi and Jan Rutten, editors, *Advanced Topics in Bisimulation and Coinduction*, volume 52 of *Cambridge Tracts in Theoretical Computer Science*, pages 290–322. Cambridge University Press, Cambridge, 2011.
16. Regivan Santiago, Manuel A. Martins, and Daniel Figueiredo. Introducing fuzzy reactive graphs: a simple application on biology. *Soft Computing*, 25(9):6759–6774, 2021.
17. Roberto Segala. *Modeling and verification of randomized distributed real-time systems*. PhD thesis, Massachusetts Institute of Technology, Cambridge, MA, USA, 1995.
18. Afsaneh Shirazi and Eyal Amir. Probabilistic modal logic. In *Proceedings of the Twenty-Second AAAI Conference on Artificial Intelligence, July 22-26, 2007*, pages 489–495. AAAI Press, 2007.
19. Franck van Breugel and James Worrell. An algorithm for quantitative verification of probabilistic transition systems. In *CONCUR 2001 — Concurrency Theory*, pages 336–350, Berlin, Heidelberg, 2001. Springer Berlin Heidelberg.
20. Franck van Breugel and James Worrell. Towards quantitative verification of probabilistic transition systems. In Fernando Orejas, Paul G. Spirakis, and Jan van Leeuwen, editors, *Automata, Languages and Programming*, pages 421–432, Berlin, Heidelberg, 2001. Springer Berlin Heidelberg.